

USU Operation System

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Introduction

The subsystem developed for this coursework is the USU Operation System, which forms the core coordination platform for the national federation of university student unions (USU). Its primary role is to manage national-level operations such as publishing announcements, approving membership changes, coordinating event information, and supporting cross-university communication.

This subsystem interacts with local student union systems and mobile applications, providing centralised services and enforcing global policies across all institutions.

Quality Requirements

Category	ID	Requirement
Security	QR-S1	All system functions must require secure authentication, enforcing role-based access (USU Officer, Union Officer, Scheduler).
	QR-S2	All communications between components (APIs, Data Store, Notification Adapter) must be encrypted using TLS 1.2+.
	QR-S3	All APIs (EventManagerAPI, MembershipAPI, AnnouncementAPI, ExternalUSUAPI) must enforce access control rules aligned with the Use Case diagram.
	QR-S4	All critical operations (event creation, announcements, membership actions, analytics access) must generate audit logs.
Performance	QR-P1	User operations through the Web Portal (e.g., Submit Event Form) must respond within 2 seconds under normal load.
	QR-P2	Component-to-component calls (e.g., Event Management → Data Store) must complete within 500 ms.
	QR-P3	Background or batch operations (announcement scheduling, analytics queries) must finish within 5 seconds.
Availability	QR-A1	The USU Operation System must maintain 99.5% uptime.
	QR-A2	Core components (Event Management, Announcement Service, Data Store) must be deployed redundantly to avoid single points of failure.
Reliability & Fault Tolerance	QR-R1	Critical operations must be atomic: if any step fails, the entire transaction is rolled back.
	QR-R2	Validation errors (as shown in the Activity Diagram) must be returned within 1 second, with meaningful messages.
	QR-R3	Notification Adapter must retry failed deliveries up to 3 times using exponential backoff.

Usability	QR-U1	The USU Officer Web Portal must follow WCAG 2.1 AA accessibility standards.
	QR-U2	All user-facing interfaces must maintain consistent layout, labels, and component placement.
Maintainability	QR-M1	The system must follow the modular architecture represented in the Component Diagram.
	QR-M2	All interactions must use standardised interfaces (EventManagerAPI, MembershipAPI, etc.) to allow independent replacement of components.
	QR-M3	All components must use a unified logging and error reporting structure.
Scalability	QR-SC1	The system must allow horizontal scaling of Event Management, Notification Adapter, and Data Store components.
	QR-SC2	The Analytics Service must scale to support peak usage without affecting performance.
Interoperability	QR-I1	Notification Adapter must be compatible with the External Notification Platform's API format.
	QR-I2	USU Integration API must expose endpoints usable by: Student App, Society Leader App, and SU Management System.
Data Integrity & Privacy	QR-D1	Data Store must maintain consistent and synchronized data for events, memberships, announcements, and analytics.
	QR-D2	Operational data and logs must be retained for a minimum of 12 months.
	QR-D3	All personal data must comply with GDPR regulations.

Justification:

The Quality Requirements defined for the USU Operation System are directly derived from its functional behaviour and architectural structure as described through the UML Use Case, Activity, and Component Diagrams.

The **Security** requirements are justified by the system's exposure to multiple roles (USU Officer, Union Officer, Scheduler) and several external integrations (Student App, Society Leader App, SU Management System, External Notification Platform). Because the system manages sensitive information—such as national-level event data, membership status, and union operations—strong authentication, access control, encryption, and audit logging are necessary to prevent misuse and ensure accountability.

The **Performance** requirements stem from the Activity Diagram, where officers must interact with the system in real time (e.g., creating events). To avoid delays in event publication or analytics retrieval, the system must maintain low response times both at user-facing and internal service levels.

The **Availability** and **Reliability** requirements follow from the fact that the USU Operation System is the central coordination platform of the entire federation of student unions. Critical services—Event Management, Announcement Service, Notification Adapter, and Data Store—must remain operational and fault-tolerant to prevent disruption of national events and communications.

The **Usability** requirements ensure that the USU Officer Web Portal is accessible and easy to operate, as officers may have differing levels of technical proficiency. The system must therefore comply with accessibility standards and provide consistent interfaces.

The **Maintainability** and **Scalability** requirements are justified by the modular architecture shown in the Component Diagram. Each subsystem—Event Management, Membership Management, Announcement, Analytics, Notification Adapter—is designed as an independent component with clear interfaces, allowing updates, replacements, and scaling without affecting the entire system.

The **Interoperability** requirements ensure that the USU Operation System can communicate reliably with external and mobile systems via the USU Integration API and the Notification Adapter.

Finally, **Data Integrity and Privacy** requirements are justified by the critical nature of the information handled (events, membership records, analytics, notifications). Ensuring accuracy, consistency, retention, and GDPR compliance is essential for legal, operational, and reputational reasons.

Together, these Quality Requirements guarantee that the USU Operation System operates securely, efficiently, consistently, and in alignment with the architecture and behaviour defined in the UML models.

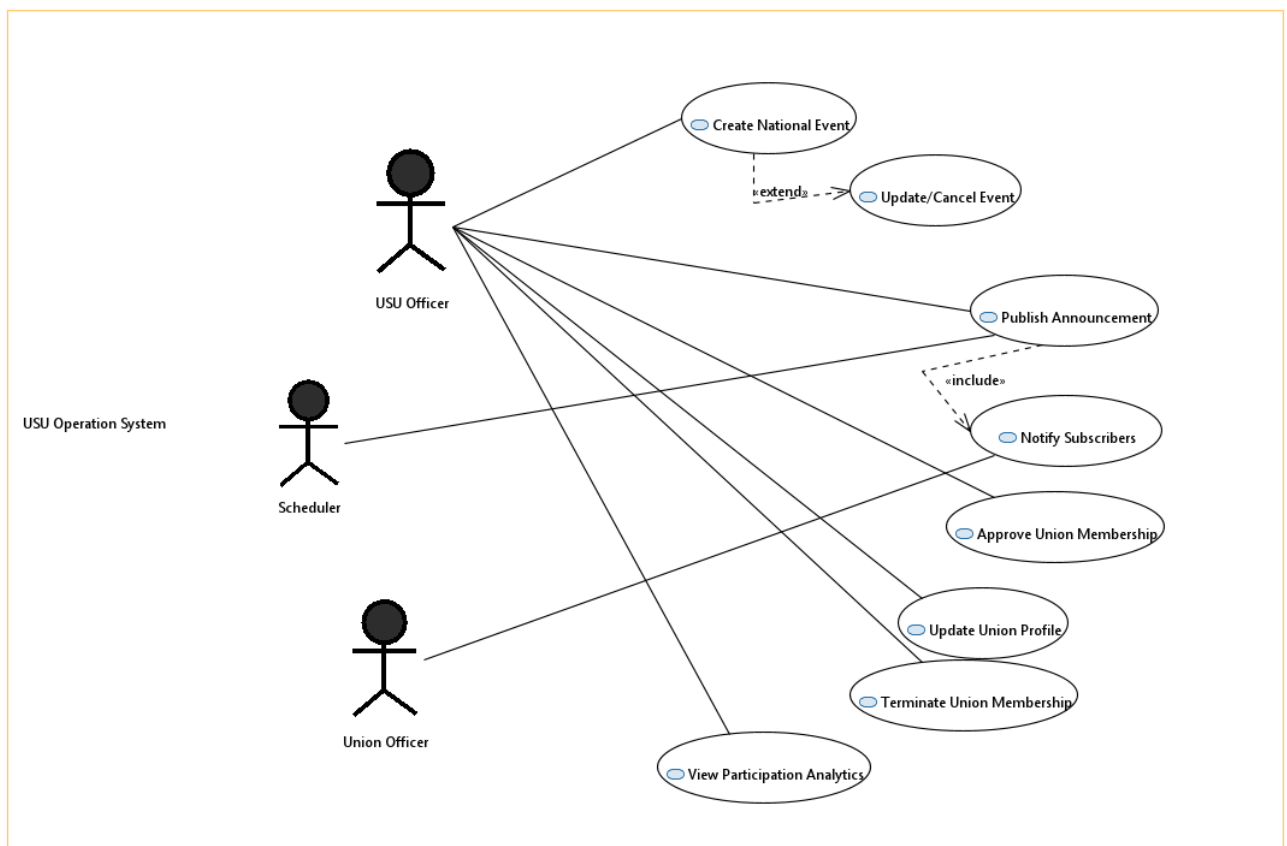
Functional Requirements

ID	Functional Requirement
FR1 — Create National Event	The system shall allow a USU Officer to create a new national event by submitting event details (title, date, description, location, target unions) using the Web Portal.
FR2 — Validate Event Data	The system shall validate all event fields (date format, missing fields, invalid structure) before submission, as shown in the Activity Diagram.
FR3 — Update or Cancel Event	The system shall allow a USU Officer to update or cancel previously created events stored in the system.
FR4 — Store Event Data	The system shall save all event-related data in the USU Data Store through the Event Management Service.
FR5 — Publish Announcement	The system shall allow a USU Officer to publish national announcements to all connected student unions.
FR6 — Notify Subscribers	The system shall transmit published announcements to external applications (Student App, Society Leader App, SU Management System) using the Notification Adapter.
FR7 — Approve Union Membership	The system shall allow a USU Officer to approve new university unions into the federation via the Membership Management Service.
FR8 — Update Union Profile	The system shall allow modifications to a union's details (name, status, region, registered societies).
FR9 — Terminate Union Membership	The system shall allow the termination of a union's membership within the federation, updating its status in the Data Store.
FR10 — View Participation Analytics	The system shall provide aggregated analytics (event attendance, union activity levels, engagement metrics) through the Analytics Service.

Use Case Model

The Use Case Model defines the functional behaviour of the USU Operation System from the perspective of its primary actor, the **USU Officer**. The model identifies the key tasks that an officer can perform when interacting with the system, including the creation, modification and cancellation of national events, the publication of announcements, and the administration of university union memberships. Additionally, the model specifies the retrieval of participation analytics to support strategic decisions within the federation.

The Use Case Diagram (Figure X) shows these interactions by mapping each system capability to a corresponding use case, illustrating how the USU Officer initiates each function. This model serves as the foundation for the subsequent Activity and Component Diagrams, ensuring that all system components support the required user-facing operations.

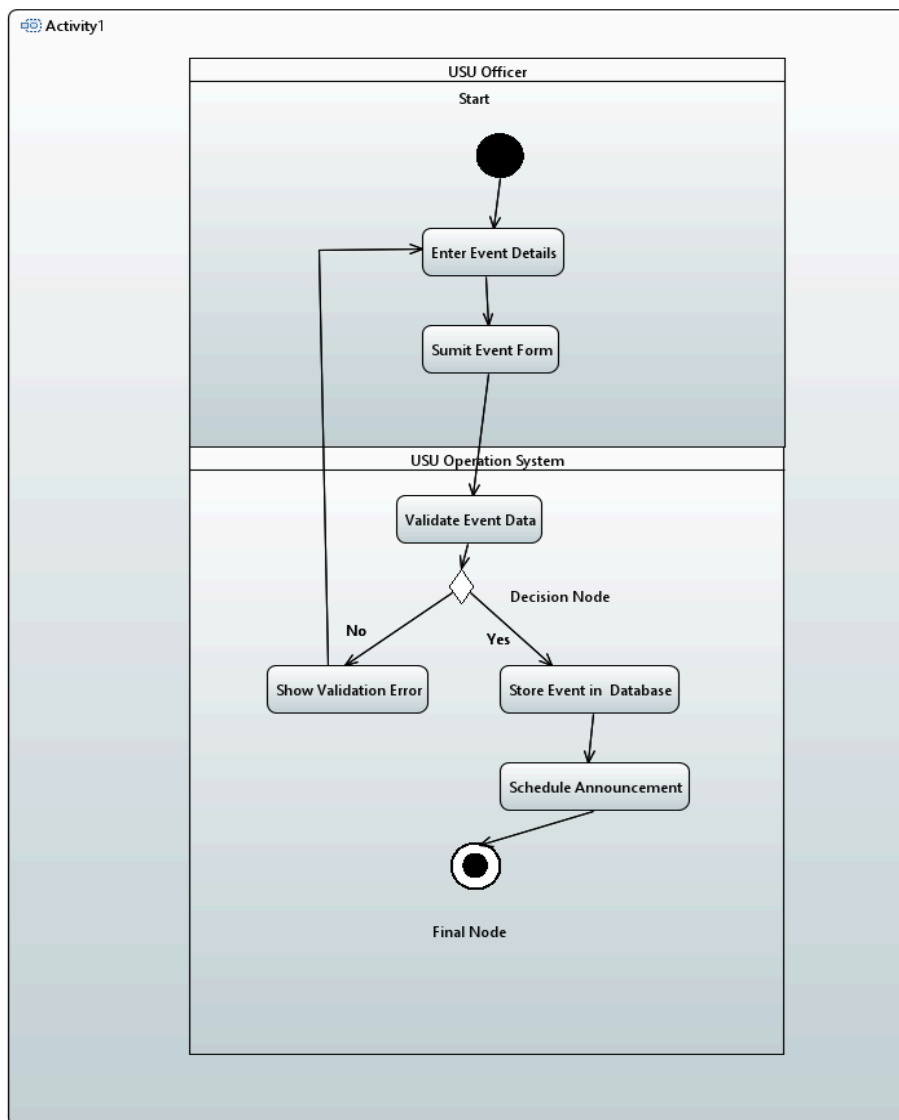


Activity Diagram

The Activity Diagram models the operational workflow of the *Create National Event* use case, illustrating how the USU Officer interacts with the system to define and register a new national event. The diagram shows the sequential flow of actions, beginning with the officer providing the event details and continuing through system-driven validation, error handling, and database storage.

The model also highlights the conditional branching required when invalid information is submitted, ensuring that the user is returned to the form with clear feedback. Once a valid event is confirmed, the system finalises the creation process and schedules subsequent actions such as announcement publication.

This diagram provides a clear behavioural view of how the system processes user input, enforces validation rules, and coordinates internal components (such as the Event Management Service and Data Store), forming an essential link between requirements and architectural design.



Component Diagram

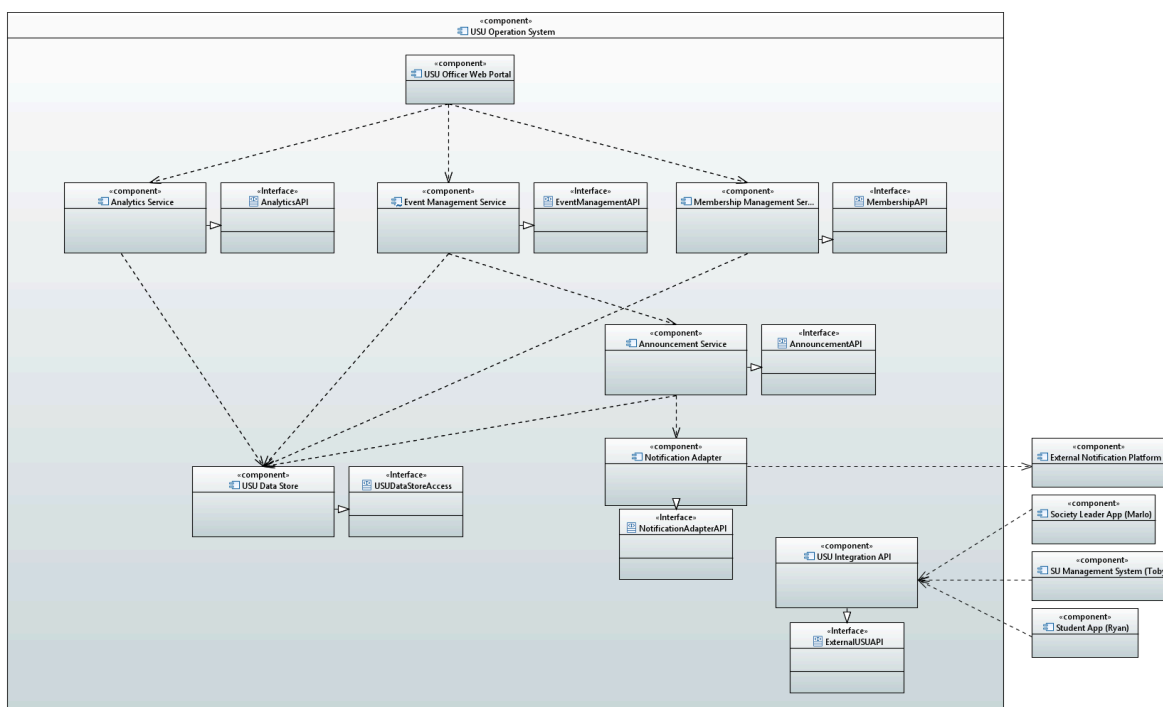
The Component Diagram provides an architectural view of the USU Operation System, showing how the subsystem is structured into independent components that collaborate to fulfil the functional requirements defined earlier. Each component encapsulates a specific set of responsibilities and exposes well-defined interfaces, enabling clear separation of concerns and supporting maintainability, scalability, and modular evolution.

The diagram identifies the major internal components of the subsystem—**Event Management Service**, **Announcement Service**, **Membership Management Service**, **Analytics Service**, **USU Data Store**, **Notification Adapter**, **USU Integration API**, and the **USU Officer Web Portal**—together forming the core architecture of the USU Operation System.

Each service provides one or more APIs (e.g., *EventManagerAPI*, *MembershipAPI*, *AnnouncementAPI*, *AnalyticsAPI*, *USUDataStoreAccess*) through interface realizations. These interfaces allow other components to interact with the service without being tightly coupled to its internal implementation.

The usage dependencies in the diagram illustrate the internal relationships between components. For instance, the **Event Management Service** depends on both the **USU Data Store** to persist event information and the **Announcement Service** to trigger communications after an event is created or updated. Similarly, the **Announcement Service** relies on the **Notification Adapter** to communicate with the external notification platform. The **USU Officer Web Portal** acts as the entry point for user interactions, invoking the main services that carry out operational tasks.

External systems—namely the **Student App**, **Society Leader App**, and **SU Management System**—communicate with the subsystem exclusively via the **USU Integration API**, ensuring secure and consistent access to national-level information across all university unions.



Contribution to the Group Work

Throughout the development of the group coursework, I contributed proactively to both the organisational and technical aspects of the project. I took responsibility for ensuring that the group remained coordinated and on track, particularly in moments where attendance or engagement from some members was inconsistent.

From the outset, I created and maintained the shared GitHub repository, structured the folders and diagrams, and produced the README file, the Members documentation and most of the minutes docs to establish a clear workflow for the team. I continuously updated the repository with my progress, including all UML models related to my subsystem, and assisted in maintaining an organised version history. I was also responsible for uploading meeting notes, integrating contributions from other members, and ensuring that the repository reflected the state of the project accurately.

In addition to my technical contributions, I took an active role in group coordination. I organised multiple WhenToMeet links to help schedule meetings, initiated discussions on Discord to maintain progress, and regularly communicated task updates and deadlines to the team. Although I was unable to attend a few sessions, I consistently ensured that my work was completed on time and that our group remained aligned with the coursework requirements.

While every member contributed at varying levels, I assumed a leading role in driving the project forward, particularly during moments of low activity from others. (19341848) contributed meaningfully during several stages of the coursework, while (19310578) participated intermittently, was responsive and present interacting during the weekly meetings. (19311988)'s involvement was minimal, in every stage of the project, only the very last week 3 days before the deadline we saw some work, not participating in meetings he attended.

Overall, my contribution to the group work included technical implementation, documentation, coordination, repository management, and acting as a consistent point of stability for the group. I ensured that the team met the necessary standards and deadlines, and that our work was submitted in a structured and coherent manner.

Conclusion

The development of the USU Operation System has provided a complete understanding of how software engineering principles translate into real system modelling, from high-level requirements to detailed architectural design. By defining functional and quality requirements, modelling user interactions through use cases and activity diagrams, and constructing a modular component-based architecture, this work demonstrates how each perspective contributes to a coherent and maintainable system.

The subsystem designed for this coursework fulfils its intended role within the wider USU ecosystem, supporting national-level coordination and communication between university unions. The diagrams and requirements developed reflect a consistent and well-structured approach that aligns user needs, system behaviour, and architectural decisions. This exercise has strengthened my ability to analyse, design, and document complex software systems following industry-standard UML practices.